Department of Computer Science and Business Systems
Academic Year 2024– 2025 (Even Semester)

Degree, Semester & Branch: IV Semester B.Tech CSBS
Course Code & Title: CS3492 & Database Management Systems
Name of the Faculty member(s): Dr.S.Erana Veerappa Dinesh, ASP/CSBS

Innovative Practice Description

- **Unit/Topic:** Unit I / SQL Fundamentals- DCL, TCL
- **Course Outcome:** CO1
- **Topic Learning Outcome:** TLO3
- **Activity Chosen:** Virtual Lab
- **Justification:**
  A Virtual Laboratory for DBMS (Database Management System) is an online platform that provides an interactive environment for learning, experimenting, and practicing database concepts without requiring a physical setup. Since SQL Fundamentals is mandatory for the database management systems, hands-on demonstration is given to the students through Virtual lab by using the following link
  https://vsit.edu.in/vlab/DBMS/Simulator.html
- **Time Allotted for the Activity:** 20 Minutes
- **Details of the Implementation:**
  The course instructor explained the concept of data control language and transaction control language in SQL fundamentals, . Figure 1 and 2 shows the Procedure for implementing the SQL Fundamentals. He opened the Virtual Lab using a fore mentioned link and simulates the queries in dbms to the students. At last 10 minutes, the course instructor chosen few students and instructed them to execute and explain the examples in virtual lab by the students shown in the fig 3.

- **CO–PO/ PSO mapping:**

CO	PO1	PO2	PO3	PSO1
CO1	3	3	2	3

(1– Low 2–Moderate 3 – High)

- **PO/ PSO mapped:**

Innovative practice	**PO1**	**PO2**	**PO3**	**PSO01**
	3	**3**	**2**	**3**
Justification for correlation	Students could be able to apply basic syntax and principles of SQL fundamentals	Students could be able to have the hands-on with implementing queries on Virtual lab	Students could be able to recollect the concept of SQL fundamentals in DBMS.	Students could able to knowledge about the SQL fundamentals to their peer group of people.

- **Images/Screenshot of the practice:**

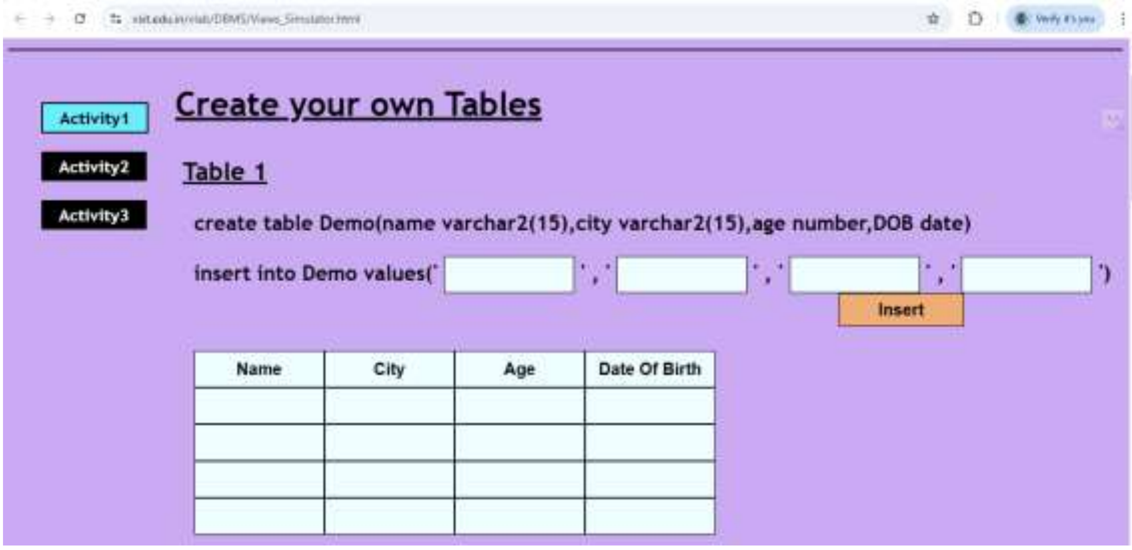

Fig1: Procedure for implementing SQL Query

Fig:2 Simulation on SQL fundamentals

Fig:3 Demonstration by the student Manoj Kumar M

- **Reflective Critique:**
  - ***Feedback of practice from students and other stakeholders:***
    - Students easily understand the data control language and transaction control language and its corresponding queries.
    - Students said that the pretest and posttest is very useful to recollect the concepts.
  - ***Benefit of the practice:*** (E.g., Outcome attainment would have increased due to innovative practice over conventional practice)
    - Students had understood the concepts of the SQL fundamentals in dbms.
    - Most of the students actively participated and enjoyed once they answered correctly in the pretest.
  - ***Challenges faced in implementation:***
    - Some students only involved in the demonstration activity.
    - The same activity will be planned in the laboratory session to make all the students familiar in the virtual environment.

References:

- https://www.ritrjpm.ac.in/images/CSBS/2023-2024/VLAB_AL3452_OS_MGM_23_24_EVEN.pdf
- https://www.ritrjpm.ac.in/images/CSBS/2023-2024/VLab_GE3151_PSPP_KU_23_24.pdf
- https://www.ritrjpm.ac.in/images/CSBS/2022-2023/SEV_GE3151_VLab.pdf
- https://www.ritrjpm.ac.in/images/CSBS/2023-2024/VL_GE3151_PSPP_BY_23_24.pdf